Lab Report: Image classification experiments

1 Introduction

In this report we compare several models on standard benchmarks. All experiments were run on a single GPU with 16 GB of memory. The learning rate was set to 0.01 with the Adam optimizer.

2 Results

Figure 1 shows the test accuracy by dataset.

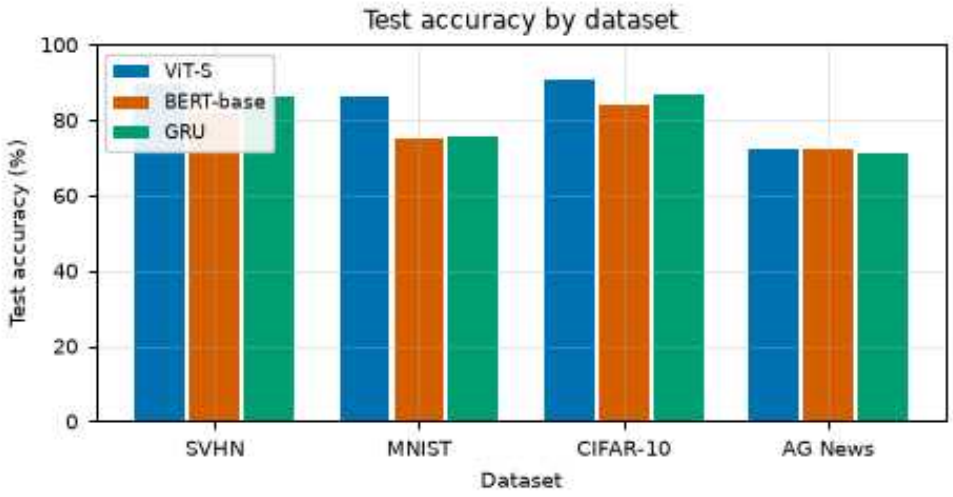


Figure 1: Test accuracy by dataset for the models in this experiment.

ViT-S outperforms GRU by 5.2 percentage points on SVHN. As shown in Figure 1, on AG News, BERT-base obtains a test accuracy of 72.0%. ViT-S reaches 87.3% test accuracy on SVHN.

Figure 2 shows the f1 score by dataset.

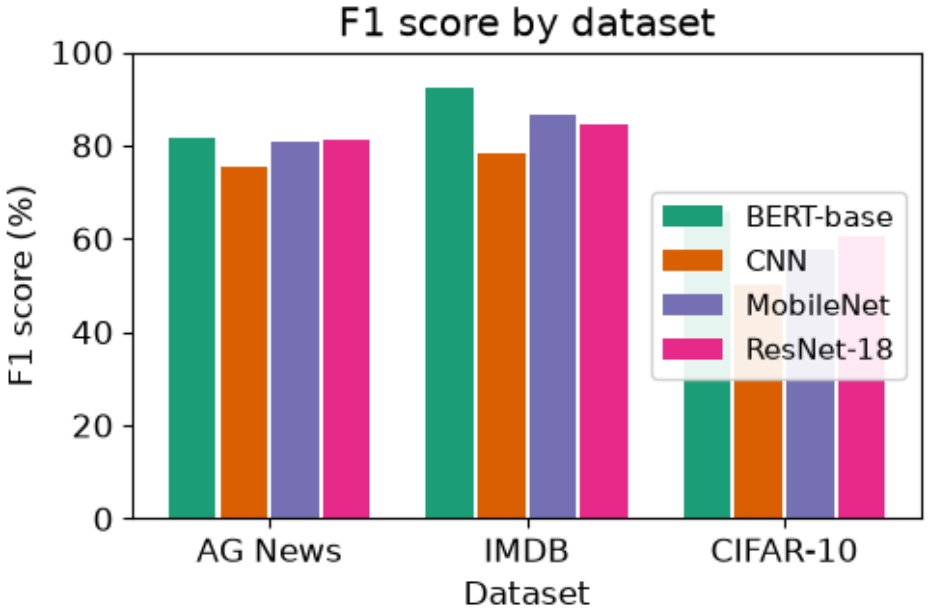


Figure 2: F1 score by dataset for the models in this experiment.

As shown in Figure 2, on AG News, MobileNet is 5.3 points better than CNN. ResNet-18 achieves 84.4% F1 score on IMDB. As shown in Figure 2, ResNet-18 reaches 110.7% F1 score on CIFAR-10. As shown in Figure 2, BERT-base outperforms MobileNet by 3.0 percentage points on AG News.

Figure 3 shows the training loss.

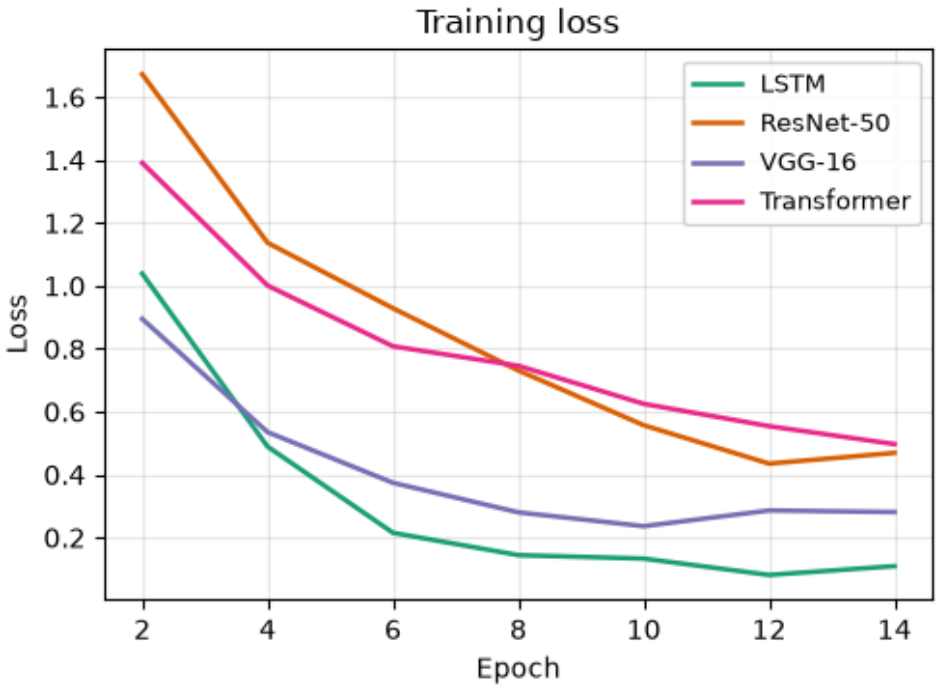


Figure 3: Training loss for the models in this experiment.

At epoch 2, the loss of LSTM is 1.039 (Figure 3). At epoch 4, the loss of VGG-16 is -0.341 (Figure 3).

Figure 4 shows the f1 score during training.

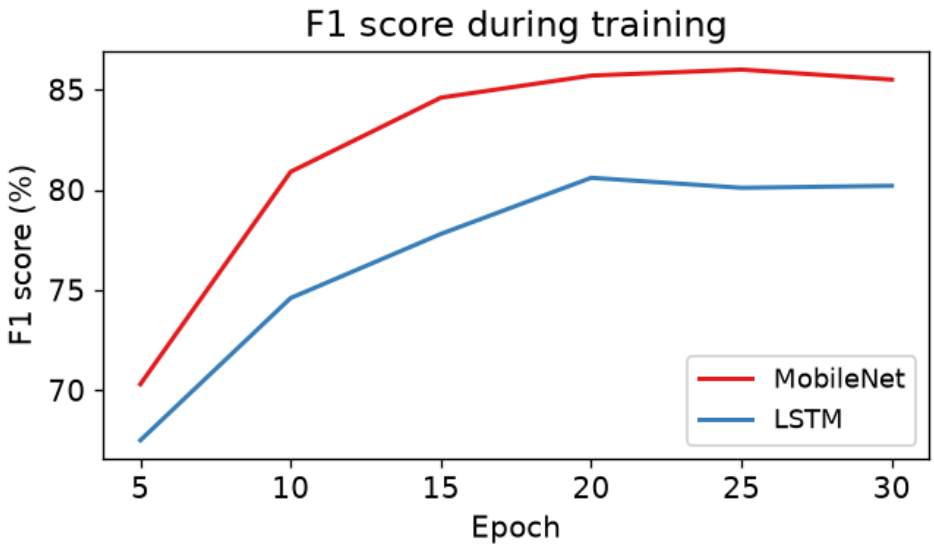


Figure 4: F1 score during training for the models in this experiment.

At epoch 25, MobileNet is 5.9 points better than LSTM. MobileNet attains a F1 score of 85.7% at epoch 20.

3 Discussion

Overall, the results are in line with what we expected from the literature.